CRYPTAD

In silico Cryptic Pocket Discovery in GWAS-Validated Alzheimer's Risk Proteins

Phase 3 Complete — Virtual Screening Results

4
Confirmed cryptic pockets

2
Target proteins

7
Final stable hits

161
MM-GBSA pairs run

Confirmed Cryptic Pockets — Recap

Phase 2 outcome: 4 pockets across 2 proteins

BIN1 BAR domain (tau propagation axis)

- site688 [PRIMARY] — inter-helix cleft, chain B membrane surface (P2Rank rank 2, score 1.99; persistence gate PASS max=0.228, 2/3 replicas)
- site680 — chain B inter-helix cleft (persistence gate PASS max=0.249)
- site1813 — peripheral site (persistence gate PASS max=0.363)

PICALM ENTH domain (Aβ endocytosis axis)

- site473 — PIP2-binding loop region (persistence gate PASS max=0.204, 1/3 replicas; borderline, P2Rank negative overridden)

All 4 pockets confirmed by: (1) metadynamics-enhanced sampling → (2) fpocket geometric detection → (3) P2Rank ML validation → (4) unbiased apo persistence gate (3 × 20 ns replicas)

Phase 3 — Virtual Screening Pipeline

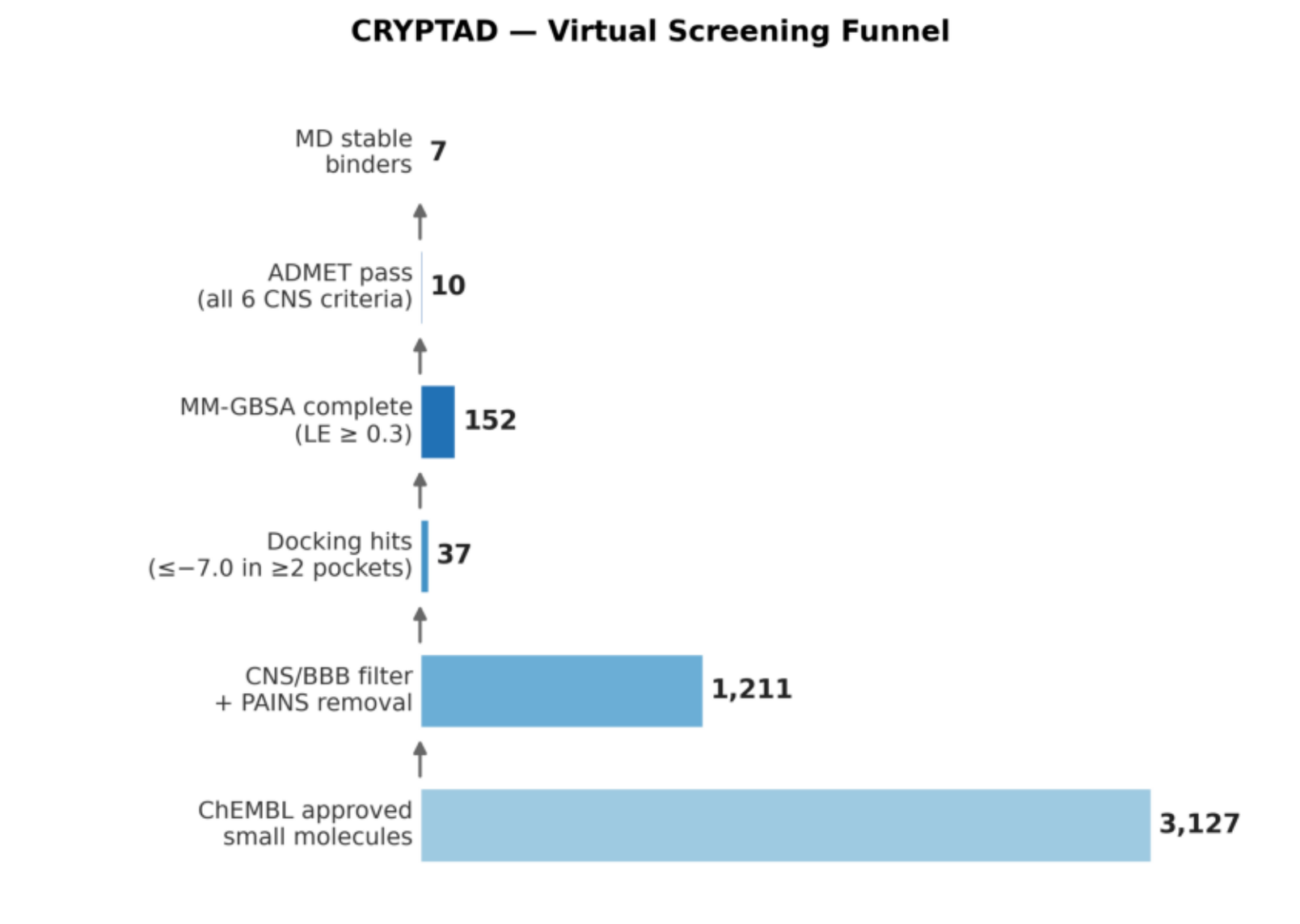
ChEMBL approved drugs → confirmed stable binders

1	2	3	4	5	6	7
Receptor prep	**Ligand library**	**Docking**	**MM-GBSA**	**LE filter**	**ADMET**	**MD stable**
pdb2pqr + Meeko 4 pocket PDBQTs	3,127 ChEMBL → CNS filter → 1,211 ready	AutoDock Vina 1.2.5 4,836 / 4,844 runs	gmx_MMPBSA GB5 161 / 200 complete	LE ≥ 0.3 152 / 161 pass	pkCSM 6 CNS criteria 10 pass	1 ns complex MD 7 confirmed

All steps run locally (receptor/ligand prep, analysis) or on TRUBA HPC (docking, MM-GBSA MD — orfoz partition, 56 cores/node)

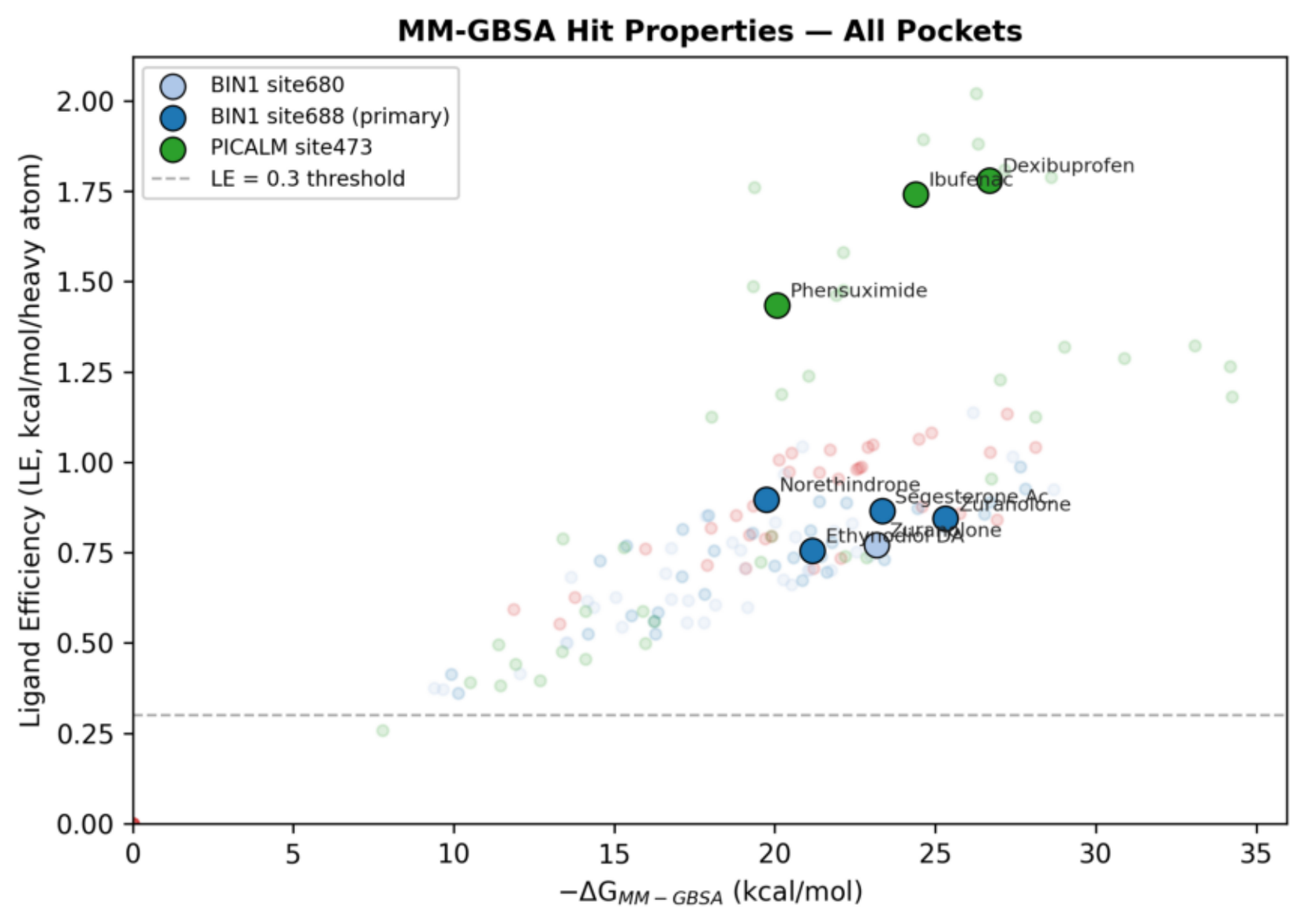
Screening Funnel

3,127 compounds → 7 confirmed stable hits



MM-GBSA Rescoring — Hit Properties

LE vs −ΔG across all 4 pockets (161 compounds)



PICALM site473 hits cluster at high LE (small, efficient binders)

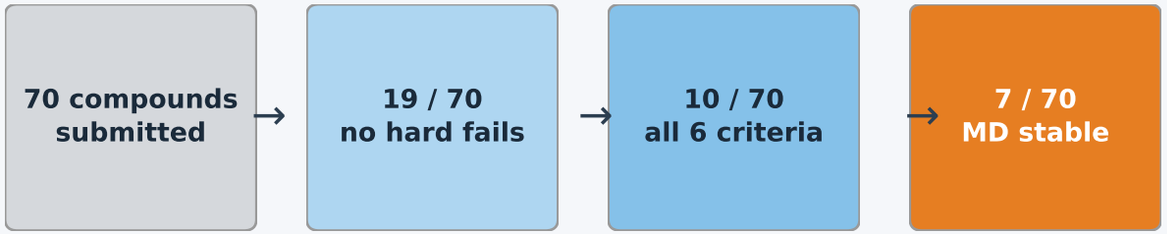
BIN1 hits at moderate LE (larger steroids)

ADMET Filtering — pkCSM Results

70 compounds screened → 10 pass all 6 CNS criteria → 7 MD-stable

Filter criteria applied:

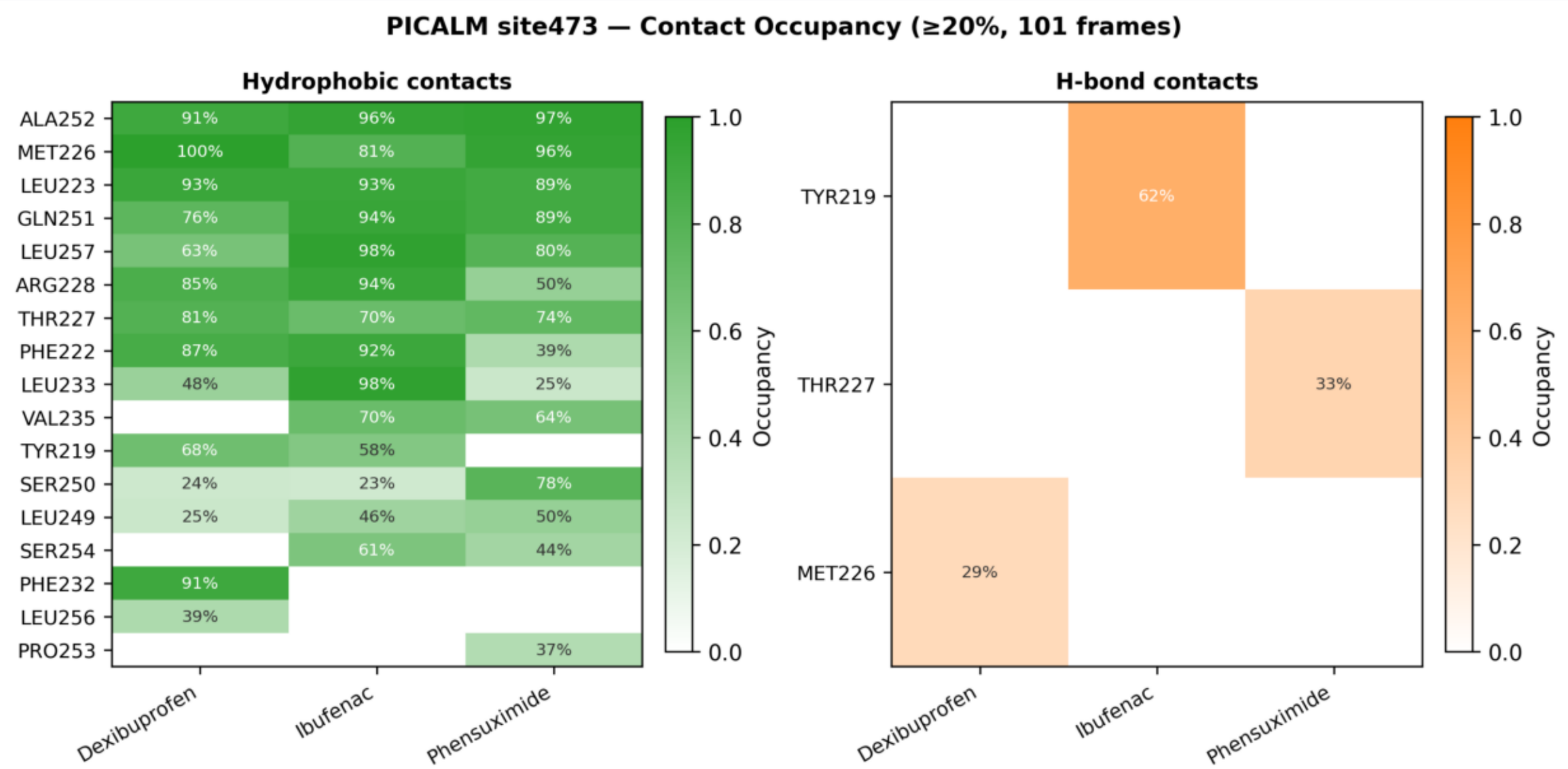
- ✓ BBB permeability LogBB > −1.0 (brain penetration)
- ✓ CNS permeability LogPS > −2.0 (CNS exposure)
- ✓ Intestinal absorption > 30%
- ✓ AMES mutagenicity = No
- ✓ hERG II cardiotoxicity = No
- ✓ Hepatotoxicity = No



MD stability check: 4 site1813 compounds (Norethindrone, Mestranol, Praziquantel, Ethinyl estradiol) dissociated from pocket during 1 ns simulation — Mestranol RMSD = 158 Å (fully unbound). Site1813 dropped from final hit list.

Binding Mode Analysis — PICALM site473

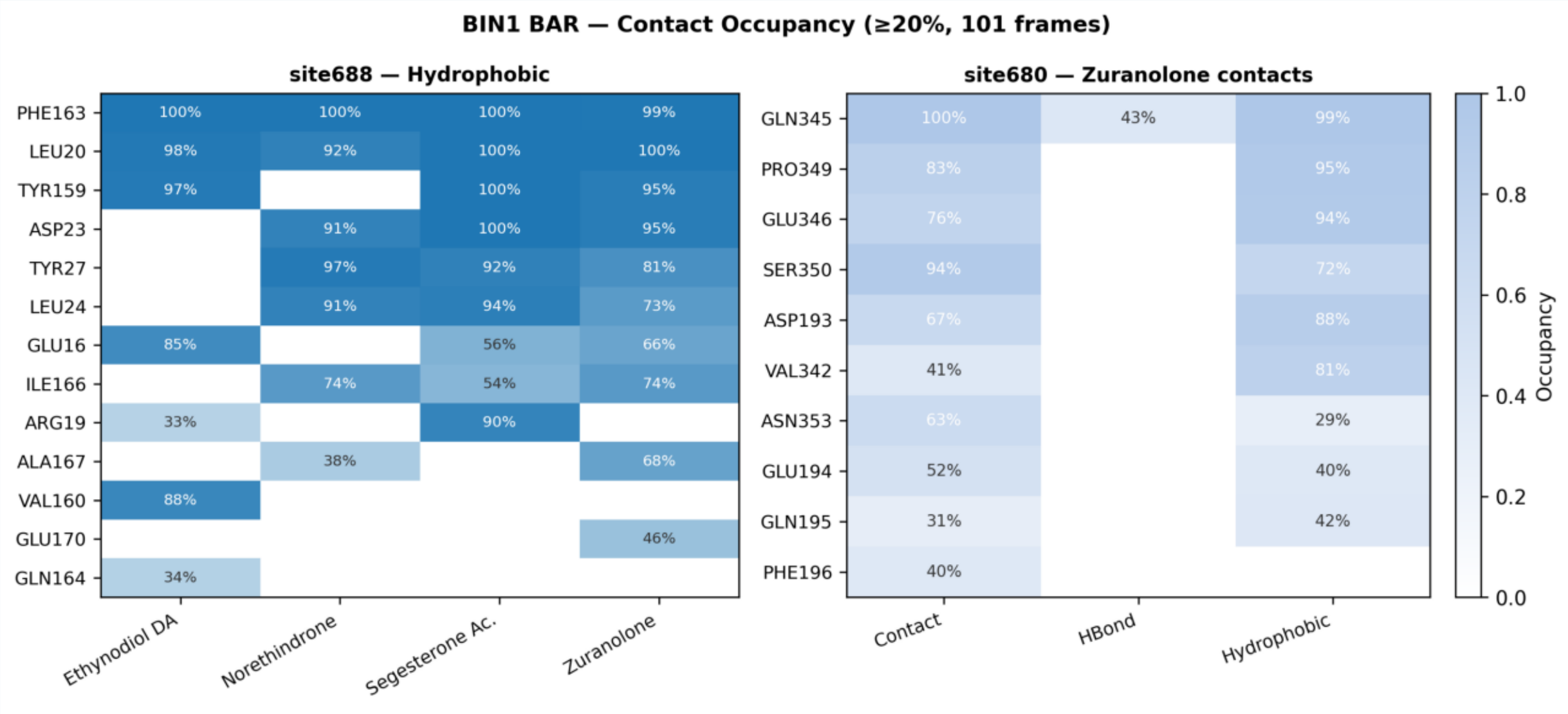
3 confirmed hits: Dexibuprofen, Ibufenac, Phensuximide



Shared hydrophobic core: ALA252 / MET226 / LEU223 (all 3 compounds, >89% occupancy) • Per-compound H-bond specificity: Ibufenac→TYR219 (62%), Phensuximide→THR227 (33%), Dexibuprofen→MET226 (29%)

Binding Mode Analysis — BIN1 BAR domain

site688: 4 steroids | site680: Zuranolone (chain B inter-helix cleft)



site688 invariant core: PHE163 / LEU20 / TYR159 / ASP23 (all 4 compounds, ≥95% occupancy) • site680 Zuranolone: GLN345 H-bond 43% + hydrophobic 95% (chain B cleft)

Final 7 Confirmed Stable Hits

Ranked by pocket and ligand efficiency | all 6 ADMET criteria satisfied

CRYPTAD — Final 7 Stable Hits: Key Properties

Pocket	Compound	dG (kcal/mol)	LE	CNS-MPO	MW	BBB (logBB)	Key H-bonds
BIN1 site680	Zuranolone	-23.2	0.772	3.3	409	0.14	GLN345 (43%)
BIN1 site688	Norethindrone	-19.7	0.897	4.9	298	0.17	none
BIN1 site688	Ethynodiol DA	-21.2	0.756	4.0	384	0.28	none
BIN1 site688	Segesterone Ac.	-23.3	0.865	4.3	370	0.07	none
BIN1 site688	Zuranolone	-25.3	0.844	3.3	409	0.14	none
PICALM site473	Dexibuprofen	-26.7	1.780	5.3	206	0.26	MET226 (29%)
PICALM site473	Ibufenac	-24.4	1.742	5.6	192	0.24	TYR219 (62%)
PICALM site473	Phensuximide	-20.1	1.435	6.0	189	-0.05	THR227 (33%)

PICALM site473
BIN1 site688
BIN1 site680

Lead Compound — Zuranolone

Dual BIN1 pocket hit | FDA-approved 2023 | CNS-active neuroactive steroid

Why Zuranolone stands out:

- **Dual-pocket binder:**
  Only hit confirmed stable at both site680 AND site688
- **FDA-approved 2023:**
  Brexanolone/Zuranolone approved for postpartum depression — already proven safe and CNS-penetrant in humans
- **Neuroactive steroid:**
  Modulates GABA-A receptors; neurological activity profile relevant to neurodegeneration
- **Strong H-bond:**
  GLN345 hydrogen bond at site680 (43% occupancy) — not just hydrophobic burial
- **BBB penetrant:**
  LogBB = +0.14 (positive — brain-penetrant by design)
- **Repurposing potential:**
  Known safety profile eliminates Phase I trials; direct path to Alzheimer's repurposing study

Zuranolone

MW	**409 Da**
LogP	**4.34**
BBB LogBB	**+0.14**
CNS LogPS	**−1.81**
LE (site688)	**0.844**
CNS-MPO	**3.26**
dG site688	**−25.3 kcal/mol**
dG site680	**−23.2 kcal/mol**
AMES	**No**
hERG II	**No**
Hepatotox.	**No**

Next Steps

Phase 4 — Complex MD validation + experimental proposal

Immediate (W15–W17):

- Complex MD — 3 × 100 ns replicas for top 3 hits: Zuranolone (site680 + site688), Ibufenac (site473), Phensuximide (site473)
- DCCM / community network analysis — identify allosteric coupling between binding site and functional interface
- Selectivity assessment — dock final hits against close structural paralogs (BIN1 SH3, CD2AP SH3) to confirm pocket specificity

Looking ahead (W17+):

- Experimental validation proposal — SPR / ITC binding assay for Zuranolone + Ibufenac against BIN1 BAR and PICALM ENTH
- Tau propagation assay — test whether site688/site680 binders reduce BIN1-driven endocytic tubulation in neuronal cell model
- Journal target: J. Chem. Inf. Model. (JCIM) or J. Med. Chem.

W14 (now)	W15	W16	W17	W18–19	W20
Figures & supv. meeting	Complex MD launch	Network analysis	MD analysis + selectivity	Write paper	Submit